

Name
Class
R. No.

• Mathematics
• IIT JEE
• Test 6(July 24 , 2010)
• Functions : Composition, Inverse(Functions)
• Maximum Marks 60
• Time 45 minutes

Instructions

1. The test contains 6 questions.
2. The marking scheme will include marks for partially correct answers also.

Section I

Q1: Let $Y = \{n^2 : n \in \mathbf{N}\} \subset \mathbf{N}$. Consider $f : \mathbf{N} \rightarrow Y$ as $f(n) = n^2$. Show that f is invertible. Find the inverse of f .

10 Marks

Q2: Let $f: \mathbf{N} \rightarrow \mathbf{R}$ be a function defined as $f(x) = 4x^2 + 12x + 15$. Show that $f: \mathbf{N} \rightarrow S$, where, S is the range of f , is invertible. Find the inverse of f

10 Marks

Q3: Prove that if $f: X \rightarrow Y$, $g: Y \rightarrow Z$ and $h: Z \rightarrow S$ are functions, then $ho(gof) = (hog) \circ f$. 10 Marks

Q4: Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be two invertible functions. Then , prove that $g \circ f$ is also invertible with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

10 Marks

Q5: Consider $f: \mathbf{R}_+ \rightarrow [4, \infty)$ given by $f(x) = x^2 + 4$. Show that f is invertible with the inverse f^{-1} of f given by $f^{-1}(y) = \sqrt{y - 4}$, where $\mathbf{R}_+$ is the set of all non-negative real numbers.

10 Marks

Q6: Consider $f: \mathbf{R}_+ \rightarrow [-5, \infty)$ given by $f(x) = 9x^2 + 6x - 5$. Show that f is invertible with
 $f^{-1}(y) = \left(\frac{(\sqrt{y+6})-1}{3} \right)$

10 Marks